HTML5 Complete Mastery Guide (A to Z)

Comprehensive full-syllabus study notes covering everything from core syntax to modern HTML5 APIs.

1. Introduction to HTML & Web Architecture

HTML (HyperText Markup Language) is the standard code used to structure a web page and its content. A web browser requests HTML files via HTTP/HTTPS, parses the tags into the Document Object Model (DOM), and renders the graphical interface.

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><title>My Page</title></head>
<body><h1>Hello World</h1></body></html>
```

- <!DOCTYPE html> informs the browser to render using modern HTML5 standards mode.
- <head> contains metadata, document title, character encoding, and external stylesheet links.
- <body> contains all visual UI content including text, images, forms, and layout containers.

2. Text Formatting & Typography (A to Z)

HTML provides specialized elements to establish text hierarchy and emphasize meaning for users and screen readers.

- <h1> to <h6>: Headings in order of semantic importance (use only one <h1> per page for SEO).
- <p>: Paragraph elements with automatic default block margins.
- & <b>: Bold text with strong semantic emphasis.
- & <i>: Italicized text with verbal stress emphasis.
- <mark>: Highlighted text for search results or callouts.
- <del> & <ins>: Strikethrough and underlined text representing edited document revisions.
- <sub> & <sup>: Subscript (e.g. H2O) and Superscript (e.g. E=mc2) elements.

3. Hyperlinks, URLs & Navigation

Links are defined using the <a> tag. Links connect web pages, files, email addresses, and specific scroll locations.

```
<a href="https://growthnest.in" target="_blank" rel="noopener noreferrer">Visit Site</a>
<a href="#section-2">Jump to Section 2</a>
<a href="mailto:info@growthnest.in">Send Email</a>
```

- href: The destination URL (absolute web URL or relative path).
- target="_blank": Opens the link in a fresh browser tab.
- rel="noopener noreferrer": Crucial security attribute to prevent tab-napping vulnerabilities.
- id scrolling: Using href="#elementId" smoothly jumps the user to that section.

4. Images, Multimedia & Embedded Content

Embedding images, responsive artwork, audio, video, and YouTube frames.

```
<img src="/assets/logo.png" alt="Company Logo" width="200" height="60" loading="lazy">
<video controls width="640">
  <source src="video.mp4" type="video/mp4">
</video>
```

- alt attribute is MANDATORY for accessibility, screen readers, and image SEO fallback.
- loading="lazy": Modern browser native lazy loading to maximize page load performance.

- <picture> tag allows serving WebP/AVIF formats with fallback to JPEG/PNG based on screen resolution.
- <iframe> embeds external content safely (YouTube, Google Maps).

5. Tables & Structured Data

Organizing 2D tabular data using semantic row and column groups.

```
<table border="1">
  <thead><tr><th>ID</th><th>Name</th><th>Status</th></tr></thead>
  <tbody><tr><td>1</td><td>Balaji</td><td>Active</td></tr></tbody>
</table>
```

- <thead>, <tbody>, and <tfoot> create clear table sections.
- <th> represents table header cells (bold and centered by default).
- colspan & rowspan: Spans cells across multiple columns or rows.

6. Semantic HTML5 Architecture

Semantic elements convey meaning to browsers, developers, and assistive screen reader technology.

- <header>: Top branding, logos, and global navigation headers.
- <nav>: Encloses primary website navigation links.
- <main>: The singular unique primary content of the document.
- <article>: Self-contained, independently distributable content (e.g., blog post, news story).
- <section>: Thematic grouping of content, typically with its own heading.
- <aside>: Secondary content (sidebars, related links, callout cards).
- <footer>: Document or section bottom info (copyright, social links, terms).

7. Complete Forms & Input Controls (A to Z)

Forms collect user data and transmit it securely to backend REST endpoints.

```
<form action="/api/register" method="POST">
  <label for="email">Email:</label>
  <input type="email" id="email" name="email" required>
  <button type="submit">Submit</button>
</form>
```

- Input Types: text, email, password, number, tel, url, search, date, time, range, color, file, checkbox, radio.
- Attributes: required, placeholder, pattern (regex), min, max, step, disabled, readonly, autofocus.
- <select> and <option>: Dropdown menus.
- <datalist>: Searchable autocomplete recommendations.
- <textarea>: Multi-line text input with configurable rows and columns.

8. HTML5 Web APIs, SEO & Best Practices

Modern HTML5 capabilities that turn websites into powerful web applications.

- SEO Meta: <meta name="viewport" content="width=device-width, initial-scale=1.0"> for responsive mobile design.
- Open Graph Tags: og:title, og:image, og:description for high-quality social sharing previews.
- Web Storage API: localStorage (permanent) and sessionStorage (browser tab session).
- <canvas> & <svg>: Rendering high performance 2D graphics and scalable vector illustrations.

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